

BANGLADESH RESEARCH AND EDUCATION NETWORK

NOC MONTHLY REPORT

July | 2021

PREPARED BY: BDREN NOC
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1. COORDINATION ISSUES

Processed 768 mail updates between BdREN NOC and member institution.

2. LINK AVAILABILITY

Table 1: Institution Wise Link Availability

Name of Institutes	Outage (Minute)	Maintenance (Minute)	Availability (w/o Maintenance)	Availability (with Maintenance)
AUB			100.00%	100.00%
AUST			100.00%	100.00%
BARI	216		99.52%	99.52%
BAU	264		99.41%	99.41%
BDU (Kaliakair Campus)			100.00%	100.00%
BDU (Dhaka Campus)	1242		97.22%	97.22%
BFRI	44		99.90%	99.90%
BLRI	157		99.65%	99.65%
BoU	216		99.52%	99.52%
BRAC			100.00%	100.00%
BRRI	1071		98.91%	98.91%
BRUR	216		99.52%	99.52%
BSMMU			100.00%	100.00%
BSMRAU	216		99.52%	99.52%
BSMRSTU	216		99.52%	99.52%
BU	216		99.52%	99.52%
BUET			100.00%	100.00%
BUP			100.00%	100.00%
BUTex			100.00%	100.00%
CoU			100.00%	100.00%
CU			100.00%	100.00%
CUET			100.00%	100.00%
CVASU			100.00%	100.00%
DIU			100.00%	100.00%
DU			100.00%	100.00%
DUET	216		99.52%	99.52%
EU			100.00%	100.00%
HSTU	216		99.52%	99.52%
ICMAB			100.00%	100.00%
IU	216		99.52%	99.52%
IUB			100.00%	100.00%
IUT	216		99.52%	99.52%
JKKNIU	1142		97.47%	97.47%
JnU			100.00%	100.00%
JU			100.00%	100.00%
JUST	216		99.52%	99.52%
KU	216		99.52%	99.52%

Name of Institutes	Outage (Minute)	Maintenance (Minute)	Availability (w/o Maintenance)	Availability (with Maintenance)
KUET	216		99.52%	99.52%
LUS			100.00%	100.00%
MBSTU			100.00%	100.00%
MIST			100.00%	100.00%
NINMAS			100.00%	100.00%
NSTU			100.00%	100.00%
NSU			100.00%	100.00%
NU			99.52%	99.52%
PSTU	216		99.52%	99.52%
PUST	216		98.62%	98.62%
RU	1021		99.52%	99.52%
RUET	216		99.52%	99.52%
SAU	216		100.00%	100.00%
SBAU			100.00%	100.00%
SUST			100.00%	100.00%

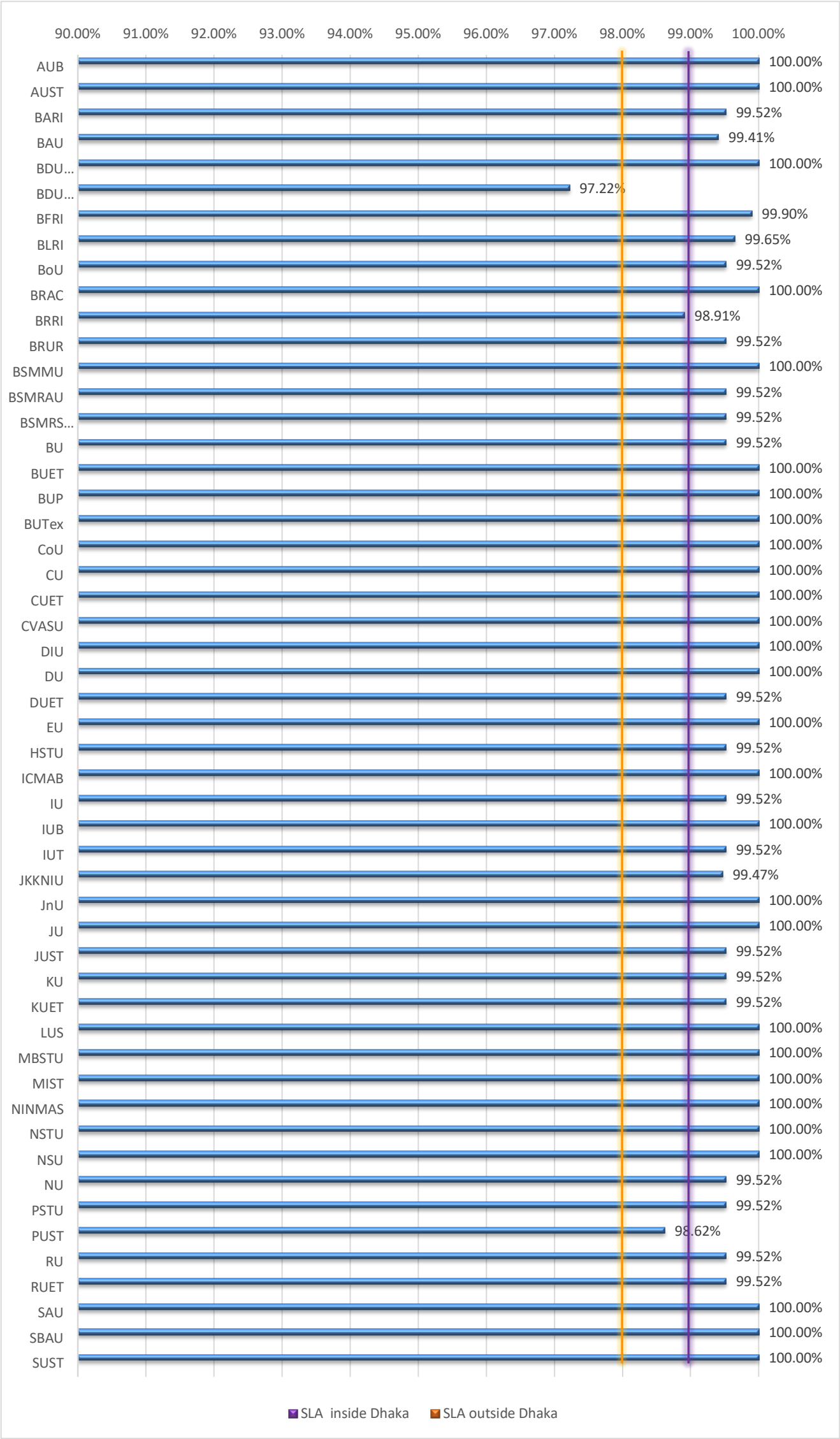


Figure 1: Institution Wise Link Availability considering “Planned Maintenance” as non-outage

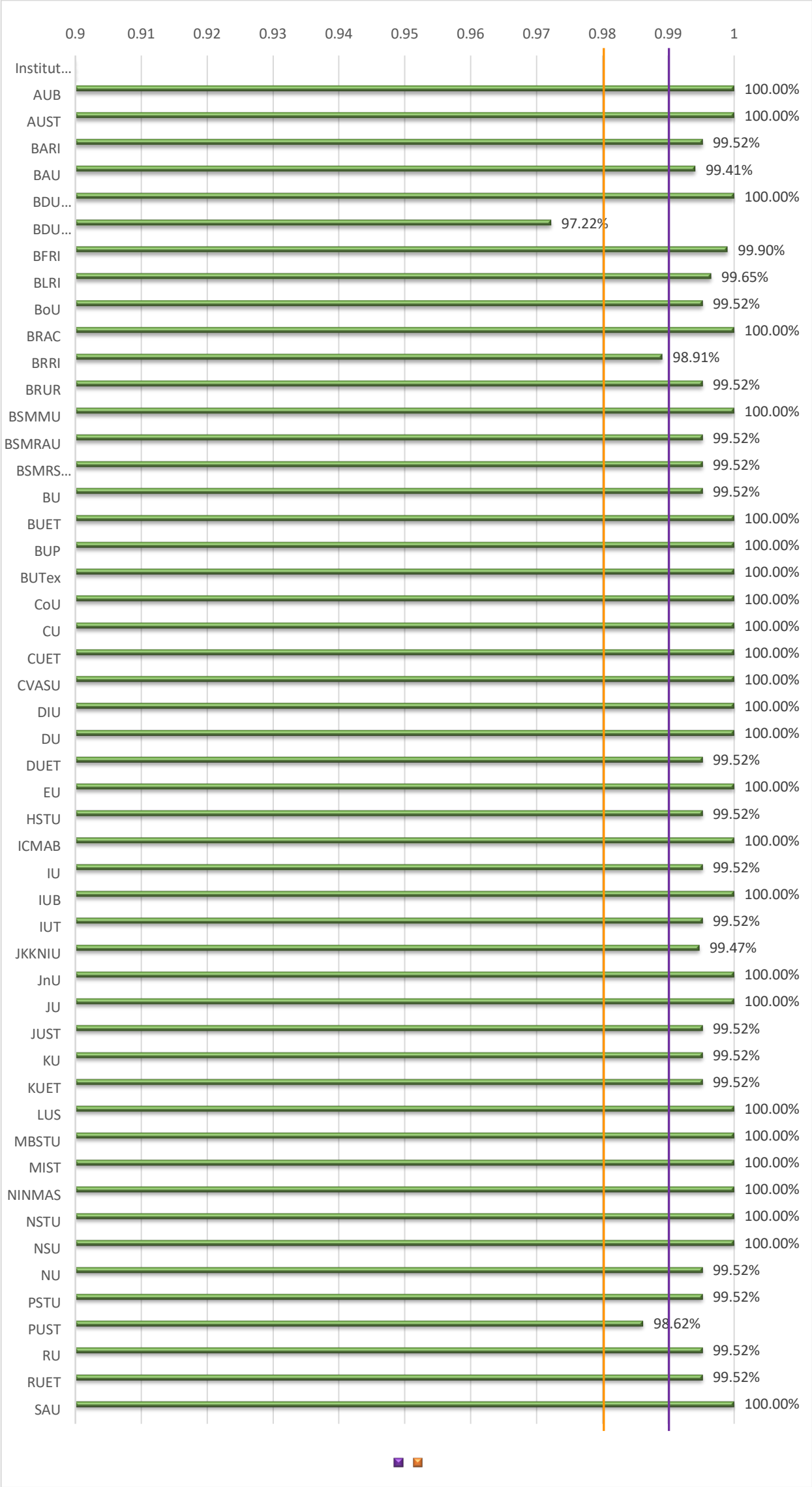


Figure 2: Institution Wise Link Availability considering “Planned Maintenance” as outage

3. MAJOR OUTAGE OF JUNE, 2021

- ❖ **Ticket#2021070301000011**
Affected Member : BRRI
Recovery Time : 270 min
SLA Compliance : Fulfilled
Problem Details : There was fiber break near BRRI main gate. BdREN Fiber team has restored the broken fiber by laying 125m overhead cable.
- ❖ **Ticket# 2021070701000013**
Affected Member : PUST
Recovery Time : 399 min
SLA Compliance : Fulfilled
Problem Details : In Natore PGCB, a patch cord was cut by rat. BdREN Fiber Team changed the patch cord and then the Tx link was getting up.
- ❖ **Ticket#2021070701000022**
Affected Member : JKKNIU
Recovery Time : 878 min
SLA Compliance : Fulfilled
Problem Details : The site was down due to fiber break by road construction work at 5.7 km away from BAU to JKKNIU in Charpara mor. BdREN team successfully restored the broken fiber.
- ❖ **Ticket# 2021070901000019**
Affected Member : BAU, BDU, BoU, BRUR, BSMRAU, BSMRSTU, BU, DUET, HSTU, IU, JKKNIU, JUST, KU, KUET, NU, PSTU, PUST, RU, RUET, BRRI and BARI.
Recovery Time : 216 min
SLA Compliance : Fulfilled
Problem Details : There were two critical Tx links down (UGC to JU and BUET to EWU). Since these links are part of the BdREN backbone, coincidentally, both links have experienced unexpected fiber breaks due to road construction work. The BUET-EWU, which is the redundant link of UGC-JU, BdREN fiber team successfully recovered the broken fiber and all of the above-mentioned sites were up within 216 min. Later on, BdREN also recovered the UGC to JU link.
- ❖ **Ticket# 20210718000001 (Mail)**
Affected Member : BDU
Recovery Time : 611 min
SLA Compliance : Fulfilled
Problem Details : Our fiber team has found that our overhear fiber has been stolen from the opposite of Prime Minister House (Near Dhaka Residential Model College. Mohammadpur). We needed 700-800m (Approx.) fiber to restore the service. Our fiber team has successfully laid the overhead fiber.
- ❖ **Ticket# 20210718000002 (Mail)**
Affected Member : BAU, JKKNIU, BFRI
Recovery Time : 44 min
SLA Compliance : Fulfilled
Problem Details : BAU Distribution was down because both primary and secondary cables were down. The secondary cable was down while BdREN team was working to restore the primary link.
BdREN Fiber team has found a fiber break on secondary link at 5500m away from MBSTU towards BDU. Then BdREN transmission team successfully reroute the circuit from another path and site became up. Later on, BdREN fiber team has successfully restored the primary fibe as well.

4. TRAFFIC STATISTICS

Table 2: Institution Wise Traffic Statistics

INSTITUTION	TOTAL TRAFFIC		AVERAGE BW		MAXIMUM BW	
	IN	OUT	IN	OUT	IN	OUT
DU	501.0 TB	106.4 TB	672.0 Mbps	118.0 Mbps	988.0 Mbps	230.0 Mbps
BUET	241.8 TB	44.6 TB	314.0 Mbps	49.0 Mbps	605.0 Mbps	125.0 Mbps
BAU	223.6 TB	35.9 TB	248.0 Mbps	48.0 Mbps	454.0 Mbps	192.0 Mbps
RU	134.0 TB	23.8 TB	169.0 Mbps	34.0 Mbps	313.0 Mbps	78.0 Mbps
JU	128.2 TB	12.6 TB	166.0 Mbps	16.0 Mbps	314.0 Mbps	55.0 Mbps
SBAU	90.4 TB	7.3 TB	110.0 Mbps	8.0 Mbps	230.0 Mbps	21.0 Mbps
CU	83.1 TB	16.3 TB	90.0 Mbps	10.0 Mbps	275.0 Mbps	60.0 Mbps
CUET	72.2 TB	6.9 TB	79.0 Mbps	9.0 Mbps	216.0 Mbps	58.0 Mbps
DUET	55.1 TB	4.5 TB	71.0 Mbps	5.0 Mbps	180.0 Mbps	32.0 Mbps
BSMMU	54.2 TB	4.6 TB	73.0 Mbps	7.0 Mbps	299.0 Mbps	31.0 Mbps
CoU	34.3 TB	2.3 TB	39.0 Mbps	2.0 Mbps	143.0 Mbps	8.0 Mbps
JUST	34.0 TB	4.6 TB	50.0 Mbps	4.0 Mbps	123.0 Mbps	39.0 Mbps
BRUR	24.0 TB	1.4 TB	31.0 Mbps	2.0 Mbps	89.0 Mbps	16.0 Mbps
RUET	23.4 TB	1.8 TB	28.0 Mbps	2.0 Mbps	92.0 Mbps	7.0 Mbps
CVASU	22.0 TB	3.3 TB	24.0 Mbps	4.0 Mbps	160.0 Mbps	20.0 Mbps
IU	22.0 TB	1.4 TB	20.0 Mbps	2.0 Mbps	54.0 Mbps	10.0 Mbps
IUT	21.9 TB	4.1 TB	105.0 Mbps	20.0 Mbps	178.0 Mbps	80.0 Mbps
JKKNIU	21.6 TB	1.6 TB	27.0 Mbps	2.0 Mbps	92.0 Mbps	11.0 Mbps
KUET	19.3 TB	2.2 TB	70.0 Mbps	8.0 Mbps	254.0 Mbps	43.0 Mbps
MBSTU	18.8 TB	2.1 TB	21.0 Mbps	4.0 Mbps	81.0 Mbps	31.0 Mbps
SUST	18.2 TB	2.4 TB	86.0 Mbps	8.0 Mbps	190.0 Mbps	30.0 Mbps
NSTU	17.3 TB	5.9 TB	30.0 Mbps	10.0 Mbps	151.0 Mbps	37.0 Mbps
JnU	13.9 TB	1.3 TB	8.0 Mbps	0.8 Mbps	39.0 Mbps	16.0 Mbps
BSMRAU	13.1 TB	1.4 TB	106.0 Mbps	8.0 Mbps	418.0 Mbps	32.0 Mbps
IUB	11.8 TB	0.7 TB	18.0 Mbps	0.9 Mbps	58.0 Mbps	7.0 Mbps
BUTex	11.4 TB	0.8 TB	14.0 Mbps	0.9 Mbps	34.0 Mbps	9.0 Mbps
NU	10.9 TB	1.3 TB	11.0 Mbps	8.0 Mbps	56.0 Mbps	97.0 Mbps
HSTU	10.4 TB	0.8 TB	13.0 Mbps	2.0 Mbps	163.0 Mbps	35.0 Mbps
PSTU	9.9 TB	0.8 TB	60.0 Mbps	4.0 Mbps	160.0 Mbps	15.0 Mbps
BLRI	9.8 TB	0.9 TB	29.0 Mbps	3.0 Mbps	99.0 Mbps	16.0 Mbps
BOU	9.3 TB	15.0 TB	16.0 Mbps	15.0 Mbps	90.0 Mbps	42.0 Mbps
BU	9.2 TB	0.5 TB	13.0 Mbps	0.7 Mbps	65.0 Mbps	20.0 Mbps
BRRRI	6.8 TB	0.5 TB	19.0 Mbps	1.9 Mbps	90.0 Mbps	9.0 Mbps
KU	5.5 TB	0.6 TB	31.0 Mbps	4.0 Mbps	88.0 Mbps	15.0 Mbps
SAU	4.9 TB	0.3 TB	24.0 Mbps	2.0 Mbps	66.0 Mbps	6.0 Mbps
BARI	3.8 TB	0.3 TB	11.0 Mbps	1.0 Mbps	100.0 Mbps	8.0 Mbps
DIU	3.6 TB	30.5 TB	11.0 Mbps	91.0 Mbps	69.0 Mbps	185.0 Mbps
BUP	3.2 TB	0.4 TB	1.0 Mbps	0.4 Mbps	15.0 Mbps	8.0 Mbps
PUST	2.9 TB	0.2 TB	9.0 Mbps	0.6 Mbps	30.0 Mbps	3.0 Mbps
ICMAB	2.7 TB	0.6 TB	7.0 Mbps	5.0 Mbps	30.0 Mbps	11.0 Mbps
LU	2.3 TB	1.0 TB	7.0 Mbps	6.0 Mbps	107.0 Mbps	27.0 Mbps
MIST	2.3 TB	0.3 TB	7.0 Mbps	0.9 Mbps	31.0 Mbps	13.0 Mbps
BSMRSTU	2.1 TB	0.1 TB	12.0 Mbps	2.0 Mbps	59.0 Mbps	24.0 Mbps
BDU	2.1 TB	2.1 TB	4.0 Mbps	2.0 Mbps	53.0 Mbps	10.0 Mbps
BFRI	1.5 TB	0.1 TB	2.3 Mbps	0.1 Mbps	20.0 Mbps	2.0 Mbps
AUST	1.0 TB	0.2 TB	3.0 Mbps	0.7 Mbps	27.0 Mbps	4.0 Mbps
EU	0.9 TB	0.3 TB	5.0 Mbps	2.0 Mbps	16.0 Mbps	8.0 Mbps
AUB	0.9 TB	0.2 TB	2.0 Mbps	0.8 Mbps	19.0 Mbps	4.0 Mbps
BRAC	0.8 TB	0.1 TB	5.0 Mbps	1.0 Mbps	28.0 Mbps	7.0 Mbps
NSU	0.0 TB	0.0 TB	0.0 Mbps	0.0 Mbps	0.0 Mbps	0.0 Mbps
VU	0.0 TB	0.0 TB	0.0 Mbps	0.0 Mbps	0.0 Mbps	0.0 Mbps
EWU	0.0 TB	0.0 TB	0.0 Mbps	0.0 Mbps	0.0 Mbps	0.0 Mbps

5. TOTAL TRAFFIC

Table 3: Total Traffic

Total Download	2017.33 TB
Total Upload	357.16 TB

6. ONLINE EDUCATION

6.1 Institution Wise vSession Usage Statistics

➤ Public Universities and Colleges:

Table 4: vSession Usage by Public Universities and Colleges

SL	University Name	Total_Class_Taken	Total_Hours	Total_Participants
1	Jahangirnagar University (JU)	3003	5410	134324
2	University of Rajshahi (RU)	2260	2759	125786
3	Jagannath University (JnU)	1840	2661	106560
4	University of Chittagong (CU)	1790	2402	99033
5	University of Dhaka (DU)	1504	2115	54779
6	Islamic University of Technology (IUT)	1429	1877	73930
7	Lalmatia Mohila College	947	650	13424
8	Bangabandhu Sheikh Mujibur Rahman Science and Technology University (BSMRSTU)	833	1142	72643
9	Bangladesh Agricultural University (BAU)	770	983	33021
10	Sher-e-Bangla Agricultural University (SBAU)	750	952	57634
11	Hajee Mohammad Danesh Science & Technology University (HSTU)	668	870	53887
12	Comilla University (CoU)	651	850	30516
13	Noakhali Science and Technology University (NSTU)	642	835	28050
14	Mawlana Bhashani Science and Technology University (MBSTU)	617	832	28605
15	Eden Mohila College	603	581	39821
16	Bangabandhu Sheikh Mujib Medical University (BSMMU)	557	809	25270
17	University of Barisal (BU)	537	695	37744
18	Bangabandhu Sheikh Mujibur Rahman Agricultural University (BSMRAU)	530	591	34086
19	Bangladesh University of Engineering & Technology (BUET)	454	586	21069
20	Cumilla Shikkha Board Govt. Model College	384	230	10563
21	Begum Rokeya University (BRU)	334	448	20503
22	Khulna University (KU)	305	413	9400
23	Islamic University, Bangladesh (IU)	305	429	17620
24	Bangladesh University of Textiles (BUTEX)	299	440	11618
25	Bangladesh Open University (BOU)	290	409	7760
26	Jashore University of Science and Technology (JUST)	286	410	7314
27	Jatiya Kabi Kazi Nazrul Islam University (JKKNIU)	284	403	15614
28	Shahjalal University of Science and Technology (SUST)	256	571	12713
29	Military Institute of Science and Technology (MIST)	241	309	8707
30	National Institute of Textile Engineering and Research (NITER)	238	259	6611

SL	University Name	Total_Class_Taken	Total_Hours	Total_Participants
31	Patuakhali Science and Technology University (PSTU)	231	268	11497
32	Bangladesh University of Professionals (BUP)	223	424	11530
33	Chittagong Veterinary and Animal Sciences University (CVASU)	206	378	12491
34	Cumilla Victoria Government College	205	228	6835
35	Pabna University of Science and Technology (PUST)	192	335	10702
36	Bangabandhu Sheikh Mujibur Rahman Maritime University	173	217	2824
37	Chittagong University of Engineering & Technology (CUET)	141	158	3976
38	Rajshahi University of Engineering & Technology (RUET)	101	117	8725
39	Rangamati Science and Technology University (RMSTU)	95	113	3774
40	Dhaka University of Engineering & Technology (DUET)	94	111	3632
41	National University, Bangladesh	84	126	2271
42	Khulna Agricultural University (KAU)	74	106	1959
43	Mymensingh Engineering College	64	85	2801
44	Khulna University of Engineering & Technology (KUET)	63	68	560
45	Bangamata Sheikh Fojilatunnesa Mujib Science & Technology University (BSFMSTU)	56	58	1862
46	Bangabandhu Sheikh Mujibur Rahman Aviation and Aerospace University (BSMRAAU)	50	74	1224
47	Sylhet Agricultural University (SAU)	43	69	3736
48	Mohammadpur Govt college, Dhaka	40	55	391
49	Sheikh Hasina University (SHU)	27	41	987
50	Chandpur Government College, Chandpur	21	16	564
51	Faridpur Engineering College	11	13	266
52	Rabindra University, Bangladesh (RUB)	4	3	235
53	Bangabandhu Digital University	3	4	85

➤
Private Universities:

Table 5: vSession Usage by Private University

SL	Institution Name	Total Class	Total Hours	Total Participants
1	Ahsanullah University of Science and Technology (AUST)	8031	9141.6182	401859
2	Green University of Bangladesh	5024	7195.3373	226788
3	Port City International University	3544	5470.6138	148254
4	Primeasia University	2921	3457.8987	84282
5	Stamford University Bangladesh	2220	3096.5163	66681
6	Leading University, Sylhet	2069	2507.4003	64644
7	Dhaka International University (DIU)	1932	2004.6503	64434
8	Shanto-Mariam University of Creative Technology	1869	2638.9006	53024
9	Asian University of Bangladesh	1785	2343.9017	48847
10	International Islamic University Chittagong (IIUC)	1780	2578.4519	52937
11	State University of Bangladesh	1565	2490.566	56132
12	City University	1540	2238.8011	77325
13	Eastern University	1500	2069.1328	67974
14	Bangladesh Army University of Engineering and Technology	1445	1437.885	42213
15	Premier University, Chittagong	1434	1773.0167	57607
16	European University of Bangladesh	1362	1499.0342	52210

SL	Institution Name	Total Class	Total Hours	Total Participants
17	Bangladesh Army International University of Science and Technology	1239	1340.3507	34370
18	International University of Business Agriculture and Technology (IUBAT)	1079	1322.185	54347
19	Northern University of Business & Technology Khulna	955	1272.6668	32409
20	North Western University (Khulna)	923	823.8986	20449
21	BGMEA University of Fashion & Technology	839	1127.9006	32334
22	ASA University Bangladesh	768	1091.9681	12877
23	Bangladesh Army University of Science and Technology (BAUST)	712	1505.2331	30942
24	Presidency University	682	831.1004	19416
25	BGC Trust University Bangladesh	628	938.7	30487
26	Prime University	549	660.6508	18220
27	Manarat International University	532	736.133	20941
28	University of Asia Pacific	426	595.4172	15615
29	Bangladesh University of Health Sciences	424	538.3667	10343
30	University of Science and Technology Chittagong (USTC)	423	460.3338	11343
31	Notre Dame University Bangladesh	422	515.6832	10782
32	Gono Bishwabidyalay	412	725.1174	13813
33	University of Information Technology and Sciences	411	596.1165	13687
34	Feni University	377	432.6658	8142
35	BRAC University	360	537.0997	10788
36	Bangladesh Islami University	344	435.7165	6404
37	Canadian University of Bangladesh	339	395.6501	9983
38	Bangladesh International School & College (BISC)	290	406.8673	15226
39	Southeast University	267	372.6167	11801
40	Atish Dipankar University of Science and Technology	266	309.016	21375
41	University of South Asia	231	298.0161	10219
42	Pundra University of Science & Technology	198	237.1166	5920
43	Rabindra Maitree University	197	229.0836	3796
44	CCN University of Science & Technology, Cumilla	183	277.8329	5927
45	North East University Bangladesh	151	206.5499	5852
46	Adamjee Cantonment College	144	153.4501	2703
47	United International University	119	197.967	4798
48	University of Liberal Arts Bangladesh (ULAB)	114	193.7334	4805
49	International Standard University	96	110.7331	2121
50	The International University of Scholars	88	116.4997	3658
51	Chittagong Independent University, Chattogram - 4000	57	83.3167	1782
52	Savar Cantonment Public School & College (SCPSC)	40	28.4332	1344
53	North South University	40	57.7835	743
54	East West University (EWU)	38	68.5666	1238
55	Bangladesh University	31	44.2666	459
56	Rajshahi Cantonment Public School and College (RCPSC)	25	6.3165	160
57	University of Liberal Arts Bangladesh	24	38.9333	1552
58	Daffodil International University	24	48.5999	5764
59	Znrf University of Management science	14	12.8501	76
60	East Delta University	10	13.6334	494
61	Notre Dame College, Dhaka	6	7.4334	307
62	Sonargaon University (SU)	6	6.7832	244
63	Ranada Prasad Shaha University (RPSU)	5	4.0834	214
64	Varendra University	2	2.4333	4
65	BdREN 3	2	0.5	2
66	Fareast International University (FIU)	2	4.5333	144

SL	Institution Name	Total Class	Total Hours	Total Participants
67	University of Creative Technology Chittagong	1	0.55	3

➤ **Medical Colleges:**

Table 6: vSession Usage by Medical Colleges

SL	Institution Name	Total Class	Total Hours	Total Participants
1	TMSS Medical College	578	666.3342	41465
2	Holy Family Red Crescent Medical College	563	562.2998	28197
3	Ibrahim Medical College, Dhaka	527	865.3834	30903
4	Faridpur Medical College	248	315.7998	22074
5	Dhaka Medical College	247	307.034	27920
6	North East Medical College	215	200.1171	19748
7	MAG Osmani Medical College, Sylhet	60	93.5665	10817
8	Kumudini Women's Medical College	23	41.4003	1796
9	Sher-e-Bangla Medical College, Barisal	6	5	357
10	Rajshahi Medical University	4	8.4833	113

➤ **Other Institution:**

Table 7: vSession Usage by Other Institution

SL	Institution Name	Total Class	Total Hours	Total Participants
1	Bangladesh Institute of Management (BIM)	307	493.9003	21374
2	Bangladesh Institute of Capital Market (BICM)	58	126.5333	1424
3	Government Organization	48	74.3333	902
4	Ministry of Education	31	40.8502	2438
5	Shaikh Burhanuddin Post Graduate College	22	22.8668	348
6	National Institute of Population Research & Training	11	13.4	524
7	Bangladesh Association of Women Scientist (BAWS)	10	11.3166	366
8	Bangladesh Research and Education Network (BdREN)	8	5.1668	79
9	Bangladesh Livestock Research Institute-BLRI	8	10.1669	523
10	Bangladesh Institute of Nuclear Agriculture (BINA)	7	14.1667	340
11	Institute of Chartered Secretaries of Bangladesh (ICSB)	6	8.8999	419
12	Faith Bangladesh	4	2.6	41
13	BANBEIS	2	4.6333	157
14	Bangladesh Academy for Securities Markets - BASM	1	0.7167	5

6.2 Institution Wise vSession Usage Graph

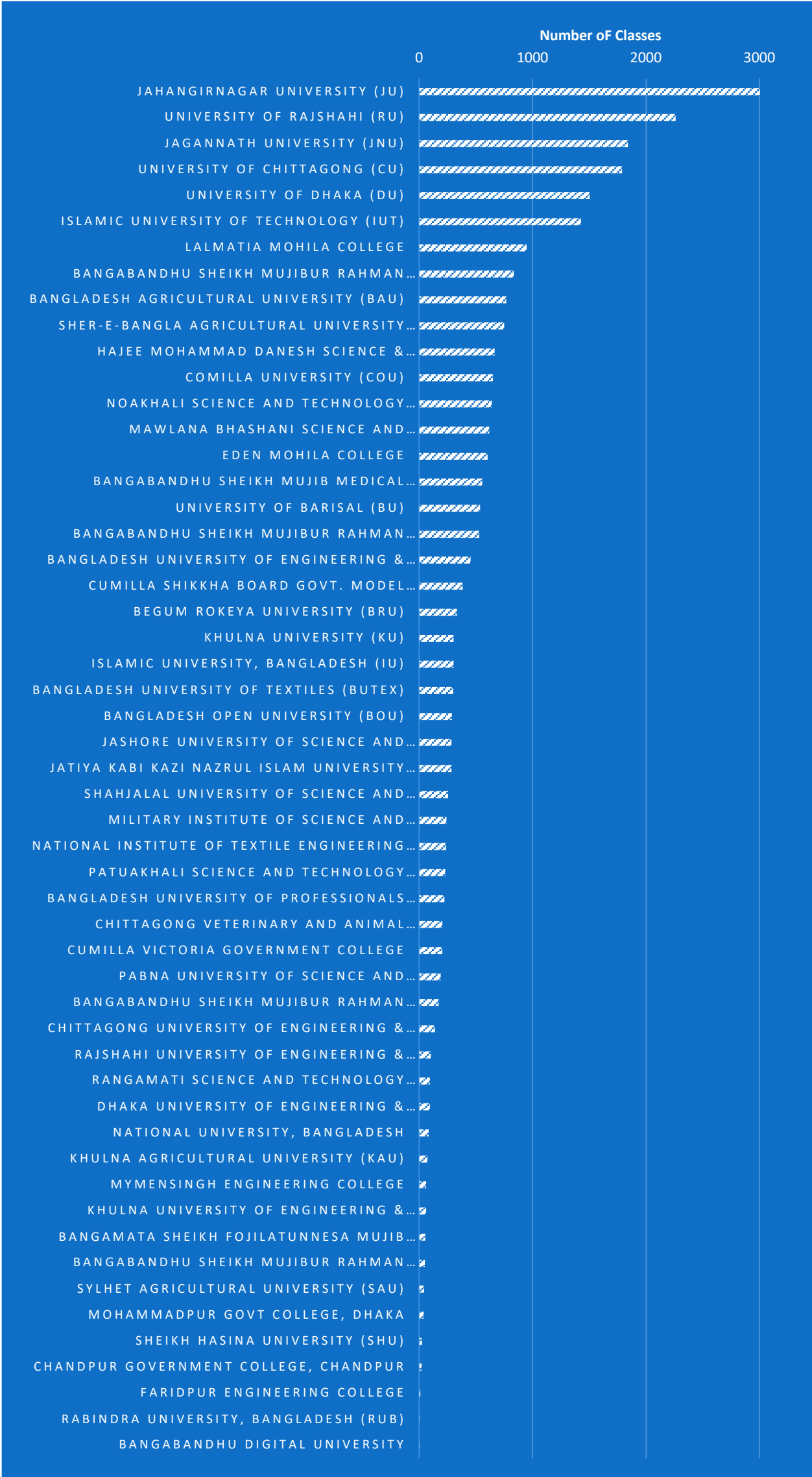


Figure 3: Public University & Colleges vSession Usage Graph

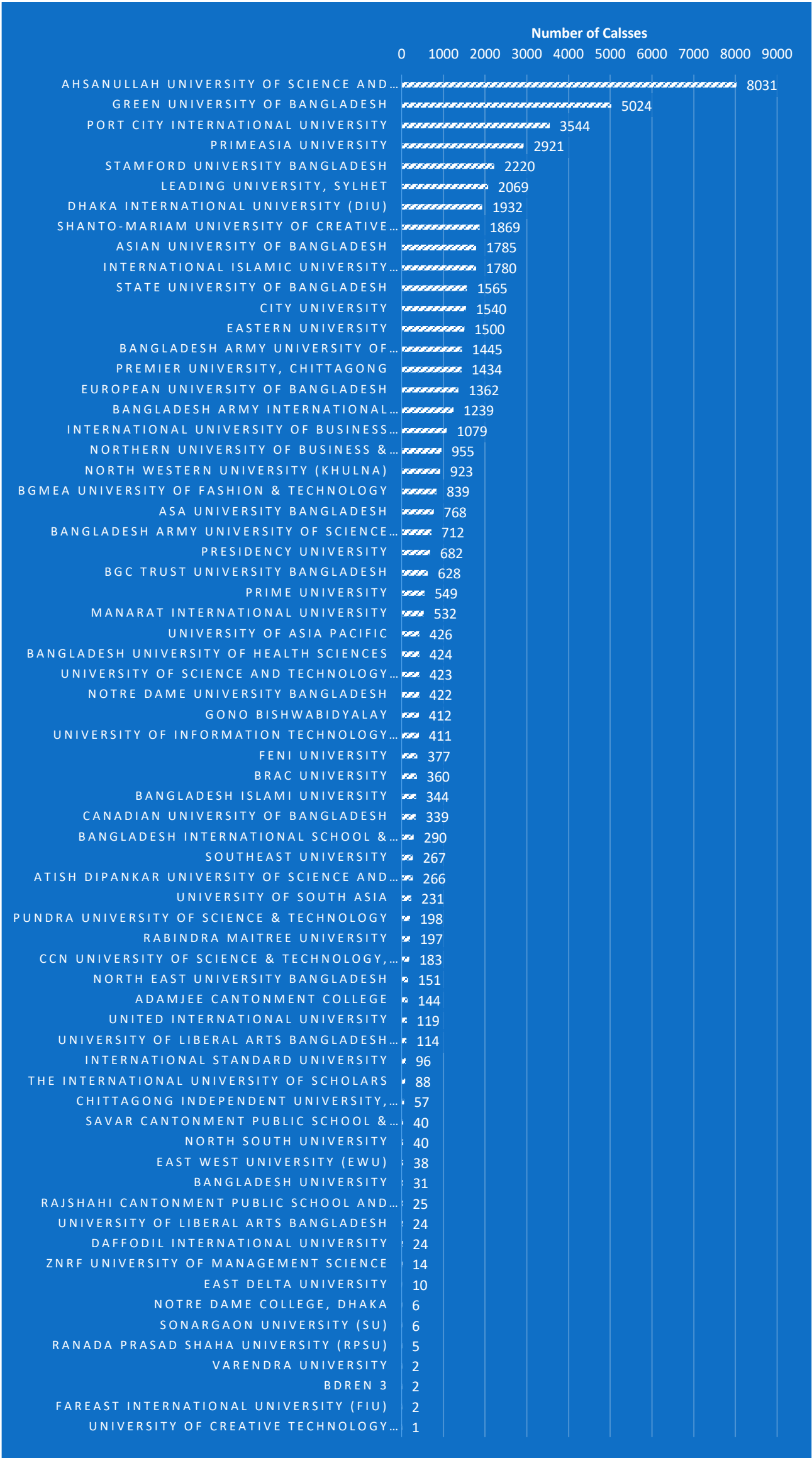


Figure 4: Private University vSession Usage Graph

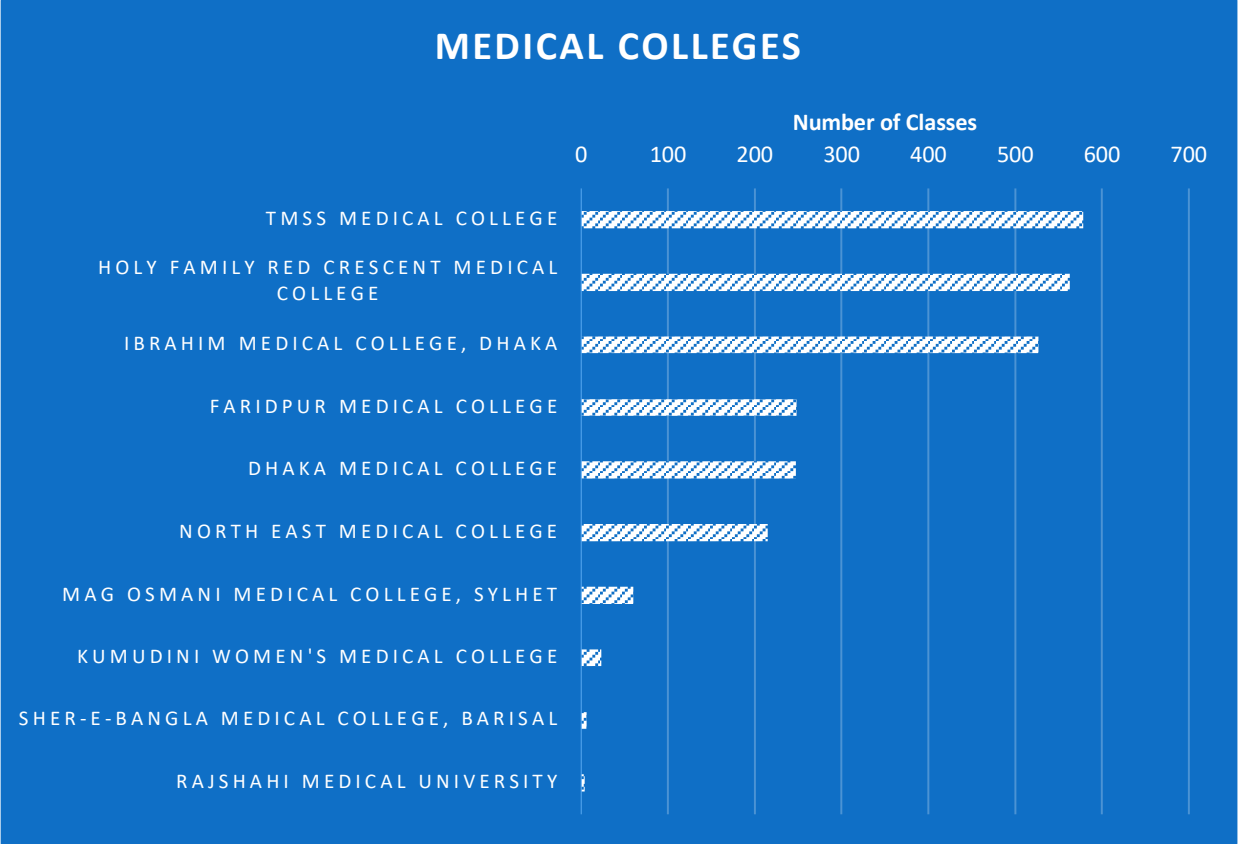


Figure 5: Medical College vSession Usage Graph

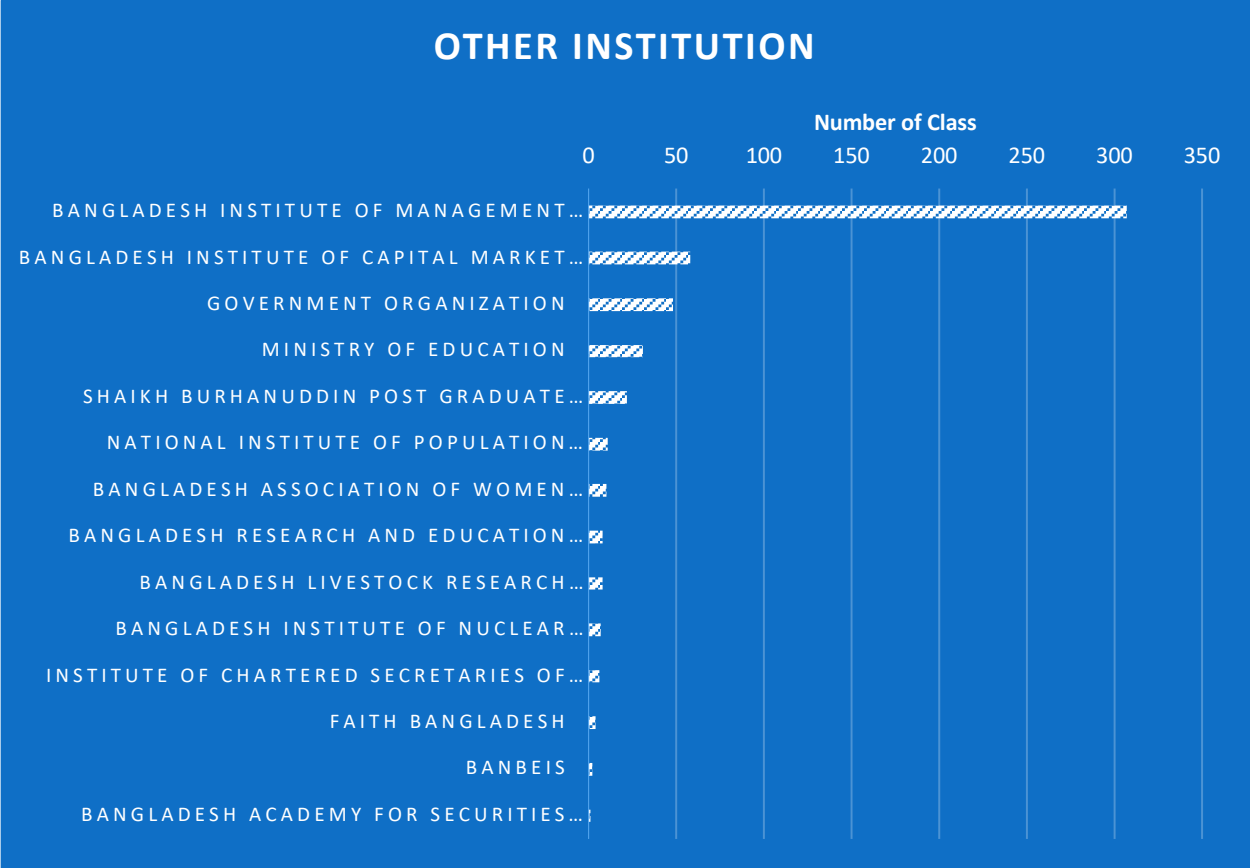


Figure 6: Other Institution vSession Usage Graph

6.3 Institution Wise On-Prem Zoom Usage Statistics

Table 8:Institution Wise On-Prem Zoom Usage Statistics

SI	Institution Name	Number Of Classes	Total Class Hours	Total Participants
1	University of Dhaka	5206	7631:39:48	238669
2	BUET	4984	7130:05:42	224846
3	Rajshahi University of Engineering and Technology	1672	2137:56:07	150406
4	United International University (UIU)	1650	2463:50:58	79845
5	Military Institute of Science and Technology (MIST)	1594	2015:01:38	62496

SI	Institution Name	Number Of Classes	Total Class Hours	Total Participants
6	Notre Dame College, Dhaka	1592	1222:20:08	410756
7	Bangladesh University of Professionals	1327	2815:10:28	103273
8	SUST	1264	1932:57:38	60056
9	CUET	895	1088:24:55	39215
10	Dhaka University of Engineering and Technology (DUET)	805	1286:15:56	40518
11	KUET	735	743:33:13	15478
12	Khulna University	530	653:00:46	11779
13	SAU	261	266:26:42	21968
14	IUT	212	316:35:00	10604
15	KWMC	188	567:44:12	48590
16	JUST	179	208:19:34	5135
17	PSTU	144	228:00:24	8432
18	BdREN	136	146:40:16	1197
19	National University	135	263:13:22	10118
20	Noakhali Science and Technology University (NSTU)	133	152:32:29	5210
21	University of Rajshahi	131	143:37:45	6976
22	Bangladesh Agricultural University	130	146:50:30	8186
23	University of Chittagong	126	170:17:07	4517
24	SPFMS	119	169:30:02	8217
25	Jahangirnagar University	113	183:46:22	7204
26	BTCL	99	164:33:41	4276
27	Asian University of Bangladesh	96	137:30:00	4398
28	JNU	92	161:09:07	2330
29	North Western University (Khulna)	92	92:11:02	2693
30	Manarat International University	85	100:39:55	4159
31	ICMAB	78	167:48:50	6835
32	Green University of Bangladesh	77	116:56:48	4100
33	Spectrum Engineering Consortium Limited	77	106:51:58	707
34	Chittagong Medical College	73	238:19:10	22324
35	Savar Cantonment Public School & College (SCPSC)	71	73:33:10	6558
36	BGC Trust University Bangladesh	67	78:45:17	1422
37	Daffodil International University	66	97:15:13	4882
38	Rajshahi Cantonment Public School and College (RCPSC)	65	63:08:09	3599
39	Sher-e-Bangla Agricultural University	64	90:43:21	11488
40	Pabna University of Science and Technology (PUST)	62	80:24:20	5079
41	Canadian University of Bangladesh	61	78:21:30	2812
42	International Standard University (ISU)	61	70:51:22	2022
43	SOMC	52	106:11:39	13531
44	ICSB	51	69:51:39	3445
45	Begum Rokeya University, Rangpur	49	76:44:21	3323
46	JKKNIU	48	105:17:40	3168
47	NIO	48	58:50:10	5311
48	UGC	47	104:27:09	7533
49	BARI	44	107:33:42	1313
50	Adamjee Cantonment College	41	35:41:07	2168
51	Rangamati Science and Technology University (RMSTU)	41	54:10:43	675
52	Ministry of Education	40	79:08:23	2045
53	Data Edge	39	74:20:32	1025
54	Bangladesh Open University	38	54:35:34	1137

SI	Institution Name	Number Of Classes	Total Class Hours	Total Participants
55	ICDDRDB	37	46:49:06	585
56	University of Liberal Arts Bangladesh (ULAB)	36	66:07:03	4403
57	Mawlana Bhashani Science and Technology Univeristy (MBSTU)	33	48:34:51	644
58	International University of Business Agriculture and Technology	32	37:47:10	1120
59	NIPORT	32	60:41:00	2816
60	Northern University of Business & Technology Khulna	32	29:36:19	1589
61	USTC	32	25:21:21	641
62	Chittagong Independent University (CIU)	30	42:03:35	1677
63	Port City international Univetsity	30	46:07:41	1868
64	Bangladesh University of Business and Technology	28	45:32:58	2308
65	Presidency University	27	39:09:09	802
66	BICM	25	64:51:55	1707
67	University of Barishal	25	40:52:46	2695
68	fDLuDCf	24	36:29:01	753
69	BGMEA University of Fashion & Technology	23	60:35:37	1905
70	BISC	22	33:38:16	2053
71	Ahsanullah University of Science and Technology (AUST)	21	22:07:49	1031
72	Sher-e-Bangla Medical College, Barisal	21	23:40:00	5261
73	Bangabandhu Sheikh Mujibur Rahman Agricultural University	20	20:57:36	804
74	Bangladesh Institute of Management (BIM)	20	20:53:29	974
75	IVTI	19	45:24:40	1639
76	University of Asia Pacific	19	23:45:06	1717
77	Dhaka Medical College	17	19:49:26	1795
78	IEDCR	17	22:24:15	371
79	Local Government Engineering Department(LGED)	17	45:59:18	1092
80	NSU	17	33:13:01	574
81	City University	16	28:09:17	1004
82	Cumilla Victoria Government College	16	34:21:06	1935
83	International Islamic University Chittagong (IIUC)	15	19:57:45	743
84	Faridpur Engineering College, Faridpur	14	24:05:56	2441
85	Bangladesh Institute of Development Studies	13	23:13:16	234
86	Islamic University, Kushtia	13	23:05:59	520
87	Eden Mohila College	12	22:07:49	404
88	Faith Bangladesh	12	16:17:11	183
89	BARD	10	15:58:16	203
90	Comilla University (CoU)	10	18:30:05	1316
91	CVASU	10	12:44:12	348
92	Dhaka Dental College	9	37:08:30	1464
93	Gono Bishwabidyalay	9	11:07:41	57
94	Independent University, Bangladesh	9	10:41:21	145
95	Shaikh Burhanuddin Post Graduate College	8	10:05:15	404
96	BRRI	7	14:30:55	365
97	BSEC	7	8:44:08	297
98	BSMRSTU	7	14:18:13	533
99	Holy Family Red Crescent Medical College	7	6:00:26	528
100	Mohammadpur Govt college, Dhaka	7	11:43:56	28
101	RMU	7	5:06:47	104

SI	Institution Name	Number Of Classes	Total Class Hours	Total Participants
102	Stamford University Bangladesh	7	14:44:40	102
103	CEDP	6	14:43:25	651
104	East West University (EWU)	6	11:36:45	167
105	Bangabandhu Sheikh Mujib Medical University	5	9:09:34	902
106	BRACU	5	4:53:17	35
107	Customs Intelligence & Investigation Directorate	4	5:31:35	272
108	Lalmatia Mohila College	4	11:09:49	2715
109	BAC	3	9:39:42	42
110	Bangladesh Fisheries Research Institute (BFRI)	2	11:14:36	214
111	BJRI	2	6:49:56	166
112	North East Medical College (NEMC)	2	3:11:22	58
113	Bangabandhu Digital University (BDU)	1	1:05:05	2
114	Department of Telecommunications	1	0:25:31	3
115	Kazla Technologies Limited	1	0:46:38	2
116	National Institute of Textile Engineering and Research (NITER)	1	1:17:06	15
117	RPSU	1	1:02:04	19

6.4 BdREN Total Zoom Usage Statistics

Table 9: BdREN Total Zoom Usage Statistics

Total Number of Meetings/Classes	111348
Total Hour of Meetings/Classes	149652
Total Number of Participants	5378990

6.5 Month-wise Number of Classes, Total Class Hours and Participants

Table 10: Month-wise Zoom Usage Statistics

Year	Number Of Classes	Total Class Hours	Total Participants
January	88,839	1,11,686	48,96,004
February	144422	1,73,143	7082885
March	111179	135669	3400250
April	115737	139956	5834405
May	79420	102116	3215695
June	127770	166413	6441336
July	111348	149652	5378990

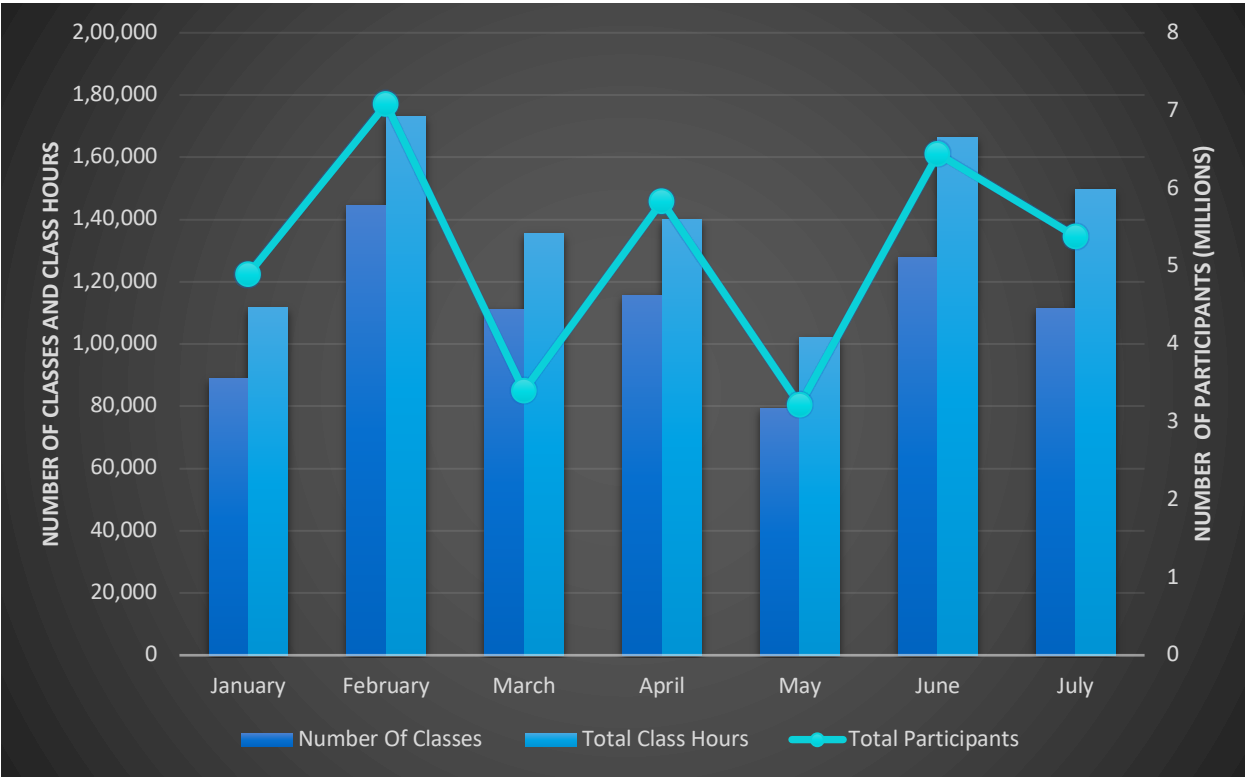


Figure 7: Month Wise Zoom Usage Graph in 2021

7. INSTITUTION WISE MAX HPC USAGE

Table 11: Institution Wise Max HPC Usage

University Name	Max CPU Usage (GHz)	Max Memory Usage (GB)
DU	592.35	913.84
IU	49.78	56.56
SUST	35.51	88.25
BUET	28.24	56.2
BU	26.09	13,35
JUST	19.25	80.23
JU	6.53	43.54
BSMRAU	1.74	24.03
JKKNIU	1.23	2.21
CVASU	1.223	18,32
PSTU	0.582	8.253
RUET	0.511	5.25
BAU	0.446	234.451
KUET	0.239	1.72
PUST	0.173	13.67
SAU	0.168	6.23
CUET	0.122	3.12
MBSTU	0.122	16.25
NITER	0.057	1.65

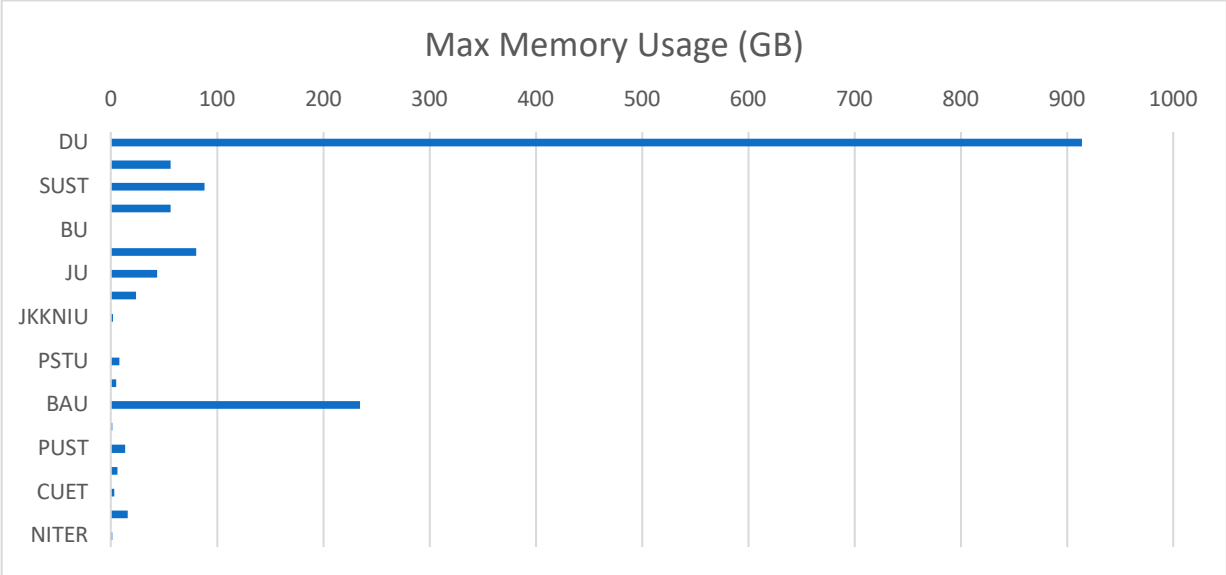


Figure 8: Institution Wise Max HPS Usage

8. ACRONYMS

AUB	Asian University of Bangladesh	JUST	Jessore University of Science & Technology
AUST	Ahsanullah University of Science & Technology	KU	Khulna University
AUW	Asian University for Women	KUET	Khulna University of Engineering & Technology
BARD	Bangladesh Academy for Rural Development	LUS	Leading University, Sylhet
BARI	Bangladesh Agricultural Research Institute	MBSTU	Mawlana Bhashani Science and Technology University
BAU	Bangladesh Agricultural University	MIU	Manarat International University
BDU	Bangabandhu Sheikh Mujibur Rahman Digital University	MMC	Mymensingh Medical College, Mymensingh
BdREN	Bangladesh Research and Education Network	NINMAS	National Institute of Nuclear Medicine & Allied Science
BFRI	Bangladesh Fisheries Research Institute	NITER	National Institute of Textile Engineering & Research
BIDS	The Bangladesh Institute of Development Studies	NKN	National Knowledge Network
BLRI	Bangladesh Livestock Research Institute	NOC	Network Operation Center
BOU	Banagladesh Open University	NREN	National Research and Education Network
BRACU	BRAC University	NSTU	Noakhali Science and Technology University
		NSU	North South University
BRRI	Bangladesh Rice Research Institute	NU	National University
BRUR	Begum Rokeya University, Rangpur	PSTU	Patuakhali Science and Technology University
BSMMU	Bangabandhu Sheikh Mujib Medical University	PUST	Pabna University of Science and Technology
BSMRAU	Bangabandhu Sheikh Mujibur Rahman Agricultural University	RAM	Random Access Memory
BSMRSTU	Bangabandhu Sheikh Mujibur Rahman Science and Technology University	RAMC	Rajshahi Medical College, Rajshahi
BTCL	Bangladesh Telecommunications Company Limited	RU	University of Rajshahi
BU	University of Barishal	RUET	Rajshahi University of Engineering & Technology
BUBT	Bangladesh University of Business and Technology	SAU	Sylhet Agricultural University
BUET	Bangladesh University of Engineering and Technology	SBAU	Sher-e-Bangla Agricultural University
BUP	Bangladesh University of Professionals	SBMC	Sher-e-Bangla Medical College, Barishal
BUTEX	Bangladesh University of Textiles	SOMC	Sylhet MAG Osmani Medical College, Sylhet
CMC	Chittagong Medical College, Chittagong	SSHMC	Shaheed Suhrawardy Medical College, Dhaka
CoU	Cumilla University	SSKMC	Sheikh Sayera Khatun Medical College, Gopalganj
CU	University of Chittagong	SSMC	Sir Salimullah Medical College, Dhaka
CUET	Chittagong University of Engineering and Technology	SUST	Shahjalal University of Science and Technology
CVASU	Chattogram Veterinary and Animal Sciences University	TB	Terabyte
DDC	Dhaka Dental College, Dhaka	UGC	University Grants Commission of Bangladesh
DIU	Daffodil International University	VU	Varendra University
DMC	Dhaka Medical College, Dhaka		
DU	University of Dhaka		
DUET	Dhaka University of Engineering and Technology		
EU	Eastern University		
EWU	East West University		
HSTU	Hajee Mohammad Danesh Science & Technology University		
ICDDR'b	International Centre for Diarrhoeal Disease Research, Bangladesh		
ICMAB	The Institute of Cost and Management Accountants of Bangladesh		
ICSB	The Institute of Chartered Secretaries of Bangladesh		
IU	Islamic University, Bangladesh		
IUB	Independent University, Bangladesh		
IUBAT	International University of Business Agriculture and Technology		
IUT	Islamic University of Technology		
JKKNIU	Jatiya Kabi Kazi Nazrul Islam University		
JnU	Jagannath University		
JU	Jahangirnagar University		

